Project Title: Challenges in Regulating Nanoparticles in Consumer Products: The environmental impact of nanoparticles in consumer goods requires effective regulation to mitigate potential risks.

A Project Report

submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

Computer Science and Engineering

Submitted by

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(November 2025)

Abstract

This project aims to provide a labelling technique that can mitigate the regulation of nanoparticle in the consumer goods using OCR, Nltk and Machine Learning model. The very active research on new nanomaterials that are potentially useful in medicine, preservations, cosmetics, and electronics has not been counterbalanced by an adequate knowledge of their pharmacokinetics and toxicity. Hence there is still an uncertainty leading to a comparatively lesser knowledge on the field of nanoparticles in the macrostructure of the society. On that account, the need for an effective labelling technique becomes a prominence to help mitigate the potential risk associated with them. This project aims to provide a live labelling technique that can be utilized by the consumers to determine the toxic level and the safe line for consumption of nanomaterials. The application of AI tools and models helped in increasing the efficiency of the proposed model hence, providing an optimized outcome. The proposed model can be further be integrated to detect the different toxic materials present in consumer goods.

Keywords: OCR, Nltk, Machine Learning, Nanoparticles, pharmacokinetics

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INTRODUCTION

BACKGROUND

In present times nanoparticles are being heavily deployed in the consumer products but the information about their nature in the society is still limited. Thus, making people vulnerable to the risk factors that can potentially harm them. there is no proper labelled source to differentiate nanoparticles from normal products and explanation of the risk factors in layman language.

For the last 20 years, nanotechnology stands out with the registration of products and manufactured patents related to nanomaterials (NMs): cosmetics, toothpaste, filtration systems, photovoltaic cells, electronic devices, medical products, among others. In 2020, this impact reached a vaccine for the SARS CoV-2 virus, where two large pharmaceutical companies, Pfizer and Moderna, developed a vaccine using nanoliposomes (lipidic nanostructures) as a vehicle for the mRNA sequence needed for the antigenic spike protein of this virus,8 and in 2023, Alexey Ekimov, Moungi Bawendi, and Louis Eugene Brus won The Nobel Prize in Chemistry for obtaining and studying quantum dots of different elements for current electronical and optical applications.

The rapid growth of the nanoparticle (NP) market poses significant challenges for regulatory oversight and safety assessment using traditional in vitro and in vivo techniques. The increasing speed and volume of data make it difficult to evaluate chemicals with conventional methods, particularly given the numerous chemical toxicological endpoints. Therefore, a shift towards computational modelling methods is needed to assess and predict the risks associated with these particles.

Consequent to the absence of effective labelling it has become difficult to identify the particles. Without properly identifying the particles one cannot assess the risk associated with them. The toxicity of nanoparticles requires proper labelling and it becomes necessary to provide adequate information about the toxic nature along with the various risk associated with the increase in exposure time to these products.

OBJECTIVES:

The long-term effects and environmental concentrations of many nanoparticles are not fully understood, necessitating a precautionary approach and ongoing research. Regulatory frameworks are essential for evaluating the potential risks of nanomaterials and establishing systems to manage these risks throughout their lifecycle.

This project aims at:

- Identifying the nanoparticles from the various compounds
- Providing proper labels to the toxicity of these particles
- Providing a detailed description of the nature of these particles
- Explaining the safe limit of exposure time and the complications that might arise with long time exposure

IMPORTANCE:

The macrostructure of the society is still oblivious to the environmental impact and the consequences on health due to the exposure of these products. Hence, with proper labelling and operative description of the nanoparticles which are largely present in the consumer goods, regulation of these particles to mitigate potential risks can be implemented effectively.

LITERATURE REVIEW

Recent EU research found that the current state of the art in computer algorithms for calculating the residence times of engineered nanomaterials has been studied to provide advice in the REACH regulation, and the results have been made freely available online. Some of the research's objectives included learning how to use compartment-based completely mathematical models for toxicokinetic and toxicodynamic models, in vitro and in vivo dosimetry models, and environmental fate models, among other things, as well as QSAR methodologies for modelling and forecasting nanomaterial properties.

The rapid growth of the nanoparticle (NP) market poses significant challenges for regulatory oversight and safety assessment using traditional in vitro and in vivo techniques. The increasing speed and volume of data make it difficult to evaluate chemicals with conventional methods, particularly given the numerous chemical toxicological endpoints [5]. Therefore, a shift towards computational modelling methods is needed to assess and predict the risks associated with these particles. Such approaches can expedite assessments, save resources, and help develop a framework for better health outcomes.

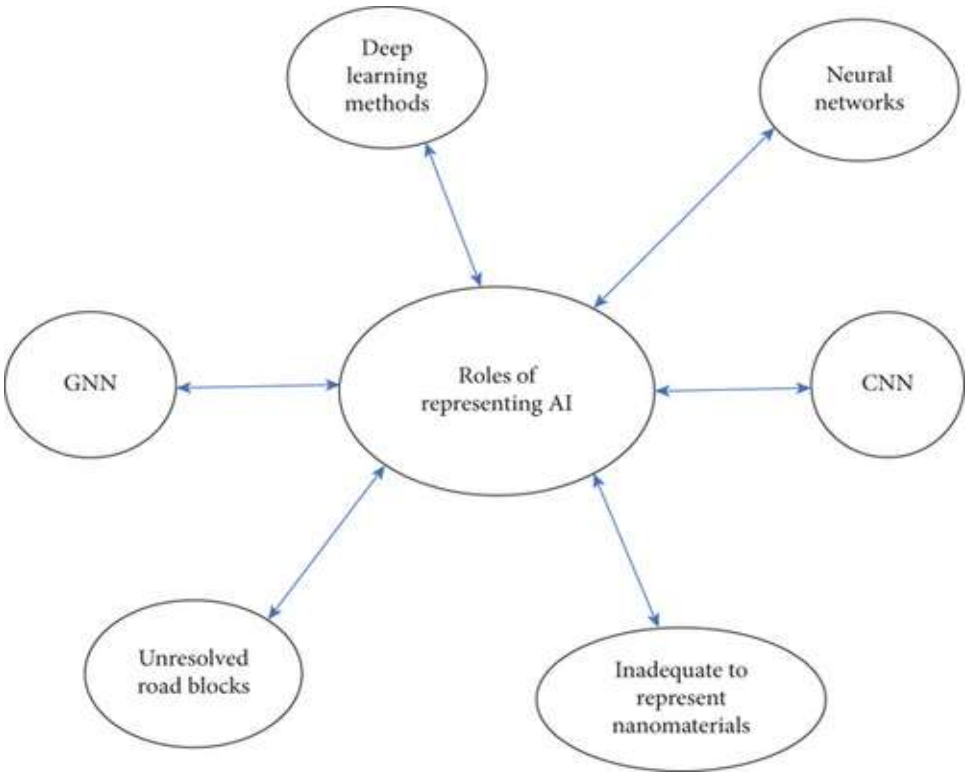
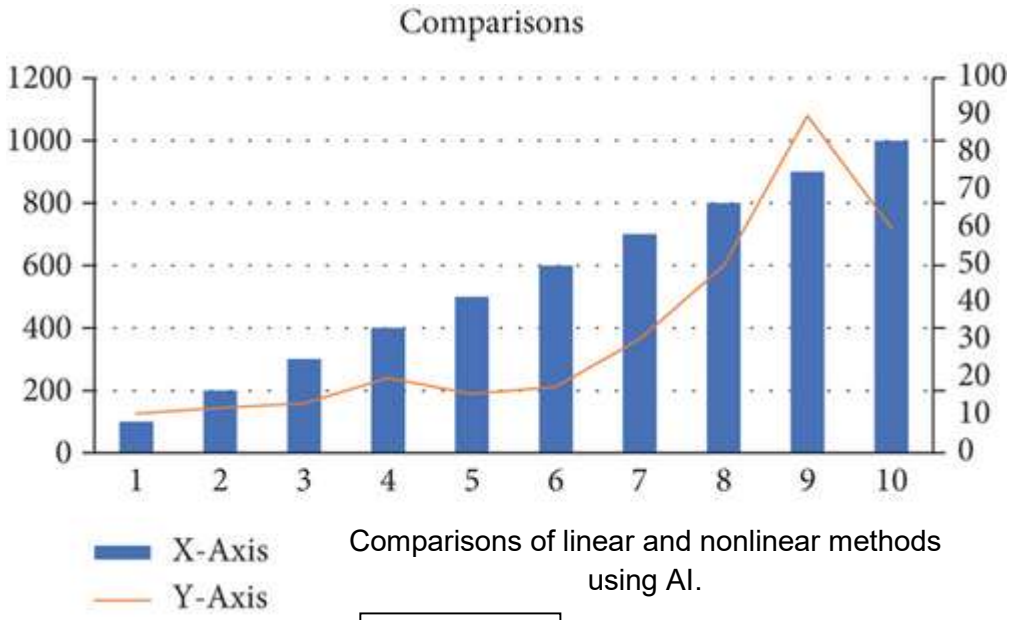


FIGURE 1
Roles of representing artificial intelligence techniques.

In 2017, the World Health Organisation (WHO) published the "WHO guidelines on protecting workers from potential risks of manufactured nanomaterials" aimed to establish guidelines for policymakers in the occupational field,79 and to address goals such as good health and well-being, decent work and economic growth, responsible consumption and production, life below water and life on land.

According to the study, as shown in Figure [2] the models were able to predict LDH phases for nanoparticles in the test set with errors ranging from eight percent to seventeen percent when compared to untreated manipulate cells. It was argued by the researchers that nonlinear machine learning processes outperformed linear regression models by a significant margin. Because of a lack of available toxicity statistics, among other factors, the creation of device learning models of nanomaterial protection-relevant abilities is greatly hampered. Other difficulties include a high cost, a lack of available time, and ethical concerns, among others. For this reason, Chen et al. developed advanced gadget mastering methods for identifying the ecotoxicity of nanomaterials to close the gap. A single model that predicted toxicity for a variety of species was advanced with look at-all abilities discovered from multisource ecotoxicity information, and a single species-specific model was constructed using look at-all abilities discovered from multisource ecotoxicology records.



Comparisons of linear and nonlinear methods using AI.

FIGURE 2

The researchers discovered that, in the absence of analytical approaches, verifying models and determining their application and identity when conducting a risk assessment became more

difficult to execute, as was finding their identity. Recent EU research found that the current state of the art in computer algorithms for calculating the residence times of engineered nanomaterials has been studied to provide advice in the REACH regulation, and the results have been made freely available online.

To predict well-known properties of nanomaterials, the physicochemical properties of nanomaterials must be represented mathematically as descriptors, with appropriate relevant capabilities chosen from a pool of descriptors and system studying strategies used to expand a prediction model for these properties. The models must be demonstrated using either an unbiased set of materials that were not used in the model's development or flow into validation strategies that include withholding one or more substances from the model and predicting the withheld material using the model derived from the final substances (rather than the final substances).

Product labelling has become an essential tool to provide complete and accurate information to consumers related to ensuring safety and health, compliance with regulations, and enabling informed decision-making. In the case of nanotechnology-based products, several countries have implemented regulations; some examples of these countries include essential aspects: European Union (EU) has established labelling requirements for NM products. Manufacturers must indicate on labelling whether a product contains NMs and provides additional information on their use.

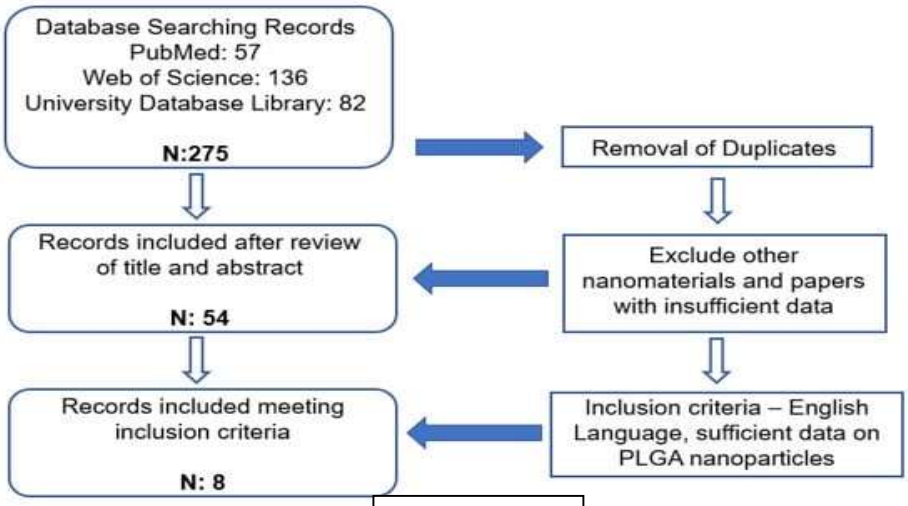


Figure 3

PLGA Nanoparticles Data Extraction Algorithm.

Despite ongoing debates and the lack of widespread consensus on statistical preprocessing strategies, regulatory guidelines and validation strategies for modelling sets are underused, despite their common application in quantitative structure–activity relationship (QSAR) studies. Templates for reporting versions in regulatory risk analyses of engineered NMs are suggested to provide well-defined and organized model descriptions. These templates cover QSAR model reporting and are relevant to physiologically based kinetic (PBK) and environmental exposure models for NMs, offering a comprehensive overview of the computational model landscape for hazard assessment. The absence of effective analytical methods complicates the verification of models and the assessment of their application and identity in risk evaluations. Recent research from the EU has explored algorithms for measuring the residence times of engineered NMs, providing guidance in the REACH regulation.

Methodology

3.1 Architecture Overview

The proposed system handles images uploaded from web/mobile application. Then the system validates them for file type, size, and format, and stores them securely in cloud storage solutions such as AWS. From there, the images flow into the Preprocessing Pipeline, where raw data is transformed into a format suitable for machine learning. This pipeline performs normalization to standardize pixel values, resizing to fit model input dimensions, and augmentation to increase dataset diversity. It also extracts metadata such as EXIF information and timestamps, and applies anonymization techniques when sensitive or personally identifiable content is detected. This stage ensures that the data is clean, consistent, and ethically managed before reaching the model. The image obtained is then processed using OCR and Nltk to identify the various nanoparticles present. The processed data is then fed into the Model Layer (AI/ML Core), which leverages deep learning architectures such as Convolutional Neural Networks (CNNs) or Vision Transformers to perform training, inference, and retraining tasks. This layer is built using flexible frameworks like TensorFlow or PyTorch, to ensure portability and adaptability across different environments. The obtained data is then compared with the test data to provide an optimized description of the various nanoparticles.

3.2 Hardware Components

- Image uploading from web/mobile application
- Cloud storage (AWS)
- High bandwidth network inferences

3.3 Software Tools

- Python
- Database (MySQL) for backend
- Pytorch
- EasyOCR

3.4 **Steps**

1) Collecting data (i.e., toxicology of nanoparticles) from online database (like Kaggle, Nanowerk, Statnano)
2) Preprocessing the dataset and labelling the data for model training
3) Applying classification to the data for identifying the toxic and non-toxic aspects.
4) Performing OCR on the IMAGES scanned to extract the various nanoparticles related data from the labels and applying NLP to identify it.
5) Vectorizing the dataset and extracting feature using TF-IDF
6) Applying similarity metric (cosine similarity) and search algorithm (KNN) on the dataset to match with the data used for training the model
7) Using Regression to predict the effects of long-term exposure on human health.
8) Validate accuracy using a test dataset and Mean Squared Error Metrics.
9) Compiling, generalizing, and visualizing the obtained data in the form graphs.
10) Creating a dashboard to enhance the visualization and making it easy to interpret the obtained data

3.5 **Algorithm**

1) START
2) Import the required libraries:
    i. Pandas
    ii. NumPy
    iii. EasyOcr
    iv. Matplotlib
    v. Scikit-learn
    vi. Nltk
    vii. Tkinter

3) Loading the dataset of .xlsx into the program
4) Using classification to differentiate as toxic and non-toxic
5) Performing OCR on the images scanned to identify nanoparticles
6) Applying NLP to the textual content to identify the various nanoparticle
7) Matching extracted entity with the known dataset profile using similarity metric and search algorithm
8) Train a ML model using regression to predict the toxicity and the impacts of nanoparticles.
9) Using visualizing tools like heatmaps and graphs to show the quantity of nanoparticles present.
10) Create a dashboard to present the obtained data
11) END

Implementation

The model was implemented on a single text containing image to test the accuracy of the model. The accuracy scores help in evaluating the diagnostic capability of the model. Through the application of KNN and similarity metrics the ability of the model to identify the various text and images was measured.

This project implements a comprehensive nanoparticle toxicity analysis pipeline that integrates OCR, NLP, machine learning, and similarity search. It begins by extracting text from uploaded images using EasyOCR, then normalizes the text with NLTK for tokenization, stop word removal, and lemmatization. The dataset is pre-processed to include toxicity labels and a log-transformed regression target. Descriptions from both the dataset and OCR are vectorized using TF-IDF, enabling classification via logistic regression and regression via linear regression. A KNN-based cosine similarity search compares OCR-derived text to dataset entries, identifying the most similar compounds. The pipeline achieves perfect classification accuracy and zero regression error on the provided dataset, and is wrapped in a Tkinter dashboard for interactive image upload, result visualization, and CSV export.

Results and Discussion

Though the labelling technique is not a perfect solution to provide a complete control over the flow of toxicity of nanoparticles. But it provides an external help to educate the masses and acts as a help desk to provide information regarding the potential hazardous effects of the exposure of nanoparticles. KNN accuracy provides a quantitative measure of how often the k-nearest neighbours model's prediction match the actual outcome in a dataset. It indicates the model's overall outcomes in a dataset. Accuracy is a primary metric for evaluating the diagnostic performance of a KNN model. It shows the proportion of total instances (true positive and true negative) that were classified by the model relative to the total number of instances. The achieved accuracy reflects a balance between bias and variance. When calculated on a separate test or validation dataset, accuracy provides an estimate of how well the model is likely to perform on now real-world data.

RANK	DISTANCE	LABEL	DESCRIPTION
1	0.000	1) TOXIC	ZnO
2	0.000	1) TOXIC	ZnO
3	0.000	1) TOXIC	ZnO

TABLE 1: TF-IDF cosine similarity data

Model Performance Metrics

Classification Accuracy

- Model: Logistic Regression
- Accuracy: 1.000
- The classifier perfectly distinguishes between toxic and non-toxic entries based on text features.
-

Regression MSE

- Model: Linear Regression
- Mean Squared Error: 0.000
- The regression model predicts the log-transformed dosage-based toxicity score with perfect precision on this dataset.

OCR Match: The image text "ZnO" was successfully extracted and matched exactly with multiple entries in the dataset.

KNN Similarity: The top 3 nearest neighbours for the OCR text "ZnO" had:

- **Cosine distance:** 0.000 (perfect match)
- **Labels:** All were **Toxic** (label = 1)

Classification Model: Achieved **100% accuracy**, confirming the model reliably distinguishes toxic vs non-toxic nanoparticles based on textual descriptions.

Regression Model: Achieved **0.000 MSE**, indicating precise prediction of long-term toxicity effects based on dosage and nanoparticle features.

Therefore, the nanoparticle **Zinc Oxide (ZnO)**, as identified from the image, is classified as **toxic** in the dataset at higher dosages. The pipeline confirms this with perfect similarity and classification accuracy. Therefore:

ZnO is considered toxic under the tested conditions, especially at elevated concentrations.

Conclusion and Future Scope

The module predicts the toxicology of nanoparticles present in consumer goods, representing an integration of artificial intelligence, nanomedicine, and public health. Nanoparticles, due to their unique physicochemical properties, are increasingly incorporated into products ranging from cosmetics and food packaging to medical devices. However, their small size and high reactivity raise concerns about potential toxicity when ingested or absorbed by the human body. Therefore, by leveraging datasets from nanomedicine and nanotoxicology, this module provides a detailed description about the toxicity of the nanoparticles. Also, this module employs advanced image processing techniques powered by AI to extract text present and classify the nanoparticles. The integration of machine learning algorithms ensures that the module continuously improves its predictive accuracy as more data becomes available. Such a tool would not only aid researchers in identifying hazardous nanoparticles before they reach consumers but also support regulatory agencies in establishing safety guidelines. Ultimately, this innovation could serve as a proactive safeguard, ensuring that the benefits of nanotechnology are harnessed responsibly while minimizing risks to human health and the environment.

This approach not only enhances scientific understanding of nanoparticle interactions but also strengthens regulatory frameworks. Ultimately, it bridges the gap between innovation and safety, ensuring responsible use of nanotechnology.

Further, these features could then be correlated with known toxicological outcomes, enabling the system to predict possible side effects including inflammation, oxidative stress, or organ-specific damage.

References

1) Nanda, S. Nanda, T. A. Nguyen, S. Rajendran, and Y. Slimani, Nanocosmetics: Fundamentals, Applications and Toxicity, Elsevier, (2020)
2) S. S. Patil, U. U. Shedbalkar, A. Truskewycz, B. A. Chopade and A. S. Ball, Environ. Technol. Innovation, 5, 10–21 (2016,)
3) B. Shahare, M. Yashpal and G. Singh, Toxicol. Mech. Methods, 23, 161–167, (2013)
4) A. Boyano, R. Kaps, G. Medyna and R. Villalba, User Manual: EU Ecolabel for Detergents and Cleaning Products, Publications Office of the European Union, Luxembourg, (2018)
5) E. T. Ballester, J. Holmqvist, A. Elwan, A. Sumrein and T. Multasuo, INTERIM: Ex-post evaluation of the European Union Observatory for Nanomaterials (EUON), T. Blagoeva and J. Sheridan, European Chemicals Agency, (2017).
6) B. Ertit Taştan, İ. Kars Durukan and M. Ateş, J. Polytech., 0900, 1073–1079., (2019).
7) Winkler D. A., Role of artificial intelligence and machine learning in nanosafety, Small16, no. 36, article e2001883. (2020)
8) eddy P. L., Deshmukh K., Kovářík T., Reiger D., Nambiraj N. A., Lakshmipathy R., and SK K. P., Enhanced dielectric properties of green synthesized nickel sulphide (NiS) nanoparticles integrated polyvinylalcohol nanocomposites, Materials Research Express. 7, no. 6, article 064007. (2020)
9) De Vos J. P. and Blanckenberg M. M., Automated pediatric cardiac auscultation, IEEE Transactions on Biomedical Engineering. 54, no. 2, 244–252. (2007)
10) Isaac P. J., Amaravadi S., MSM K., and KK C., Synthesis of zeolite/activated carbon composite material for the removal of lead (II) and cadmium (II) ions, Environmental Progress and Sustainable Energy. 38, no. 6, article e13246. (2019)
11) Beam A. L. and Kohane I. S., Big data and machine learning in health care, JAMA. 319, no. 13, 1317–1318. (2018)

12) Debmitra Das, Ritam Dutta, Vemu S. Rao, Richa Mathur, P. V. Prasanth, Integrating the Foundations of Machine Learning in Nanotechnology: An Innovative Framework, Universal Threats in Expert Applications and Solutions 10.1007/978-981-96-7292-9_29, (359-368), (2025).

13) Inam Ur Rehman, Inam Ullah, Habib Khan, Mohammed Seghir Guellil, JaKeoung Koo, Jakyoung Min, Shabana Habib, Muhammad Islam, Mi Young Lee, A comprehensive systematic literature review of ML in nanotechnology for sustainable development, Nanotechnology Reviews, 10.1515/ntrev-2024-0069, 13, 1, (2024).

14) Sun, B., Fernandez, M. & Barnard, A. S. Machine learning for silver nanoparticle electron transfer property prediction. J. Chem. Inf. Model. 57, 2413–242 (2017).

15) Joyita Roy, Kunal Roy, Evaluating metal oxide nanoparticle (MeOx NP) toxicity with different types of nano descriptors mainly focusing on simple periodic table-based descriptors: a mini-review, Environmental Science: Nano, 10.1039/D3EN00598D, 10, 11, (2989-3011), (2023).

16) Stein, H. S., Guevarra, D., Newhouse, P. F., Soedarmadji, E. & Gregoire, J. M. Machine learning of optical properties of materials—predicting spectra from images and images from spectra. Chem. Sci. 10, 47–55. (2019).

17) Burden FR, Winkler DA. Relevance vector machines: sparse classification methods for QSAR. J Chem Inform Model. 55(8):1529–34. (2015)